

**SECTION B**

Answer **all** questions.

8. The Mohs scale measures the hardness of different materials and runs from 1 to 10, with 10 being the hardest. A fingernail is rated as 2.5 on the Mohs scale so any material that can be scratched with a fingernail has a hardness of less than 2.5. Graphite and the minerals halite and tachyhydrite, can all be scratched with a fingernail.

- (a) The hardness value of graphite is approximately 1.5.

Describe the structure of graphite and explain why it is soft.

[2]

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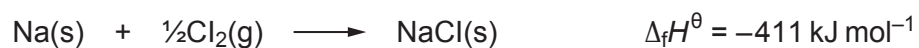
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- (b) Halite is the mineral name for rock salt. It contains sodium chloride.

The equation corresponding to the standard enthalpy change of formation of sodium chloride is as follows.



Some standard enthalpy changes are given in the table.

| Enthalpy term                     | Standard molar enthalpy change / kJ mol <sup>-1</sup> |
|-----------------------------------|-------------------------------------------------------|
| first ionisation energy of sodium | 494                                                   |
| electron affinity of chlorine     | -364                                                  |
| bond energy of Cl <sub>2</sub>    | 242                                                   |
| enthalpy of atomisation of sodium | 109                                                   |

- (i) State the enthalpy of atomisation of chlorine.

[1]

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- (ii) Use the standard enthalpy changes to find the standard enthalpy change of lattice formation of sodium chloride. [3]

Standard enthalpy change of lattice formation = .....  $\text{kJ mol}^{-1}$

- (iii) A student suggests that the entropy change for the formation of sodium chloride must be positive as the reaction occurs easily.

Is the student correct? Justify your answer. [2]

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